Big-O Notation

Prof. Timothy Sun

Big-O Notation

$$
\begin{aligned}
&\textsc{InsertionSort}(A[1,\ldots,n]):\\
&\quad \textbf{for } i = 2,\ldots,n\\
&\qquad\quad key := A[i]\\
&\qquad\quad j := i-1\\
&\qquad\quad \textbf{while } j > 0 \text{ and } A[j] > key\\
&\qquad\qquad\quad A[j+1] := A[j]\\
&\qquad\qquad\quad j := j-1\\
&\qquad\quad A[j+1] := key
\end{aligned}
$$

"Insertion sort runs in O($\boldsymbol{n}^2$) time."

Why use Big-O notation?

- Language/machine/unit-independent
- Isolating the “bottleneck”
- Captures long-term behavior
- Easier to analyze

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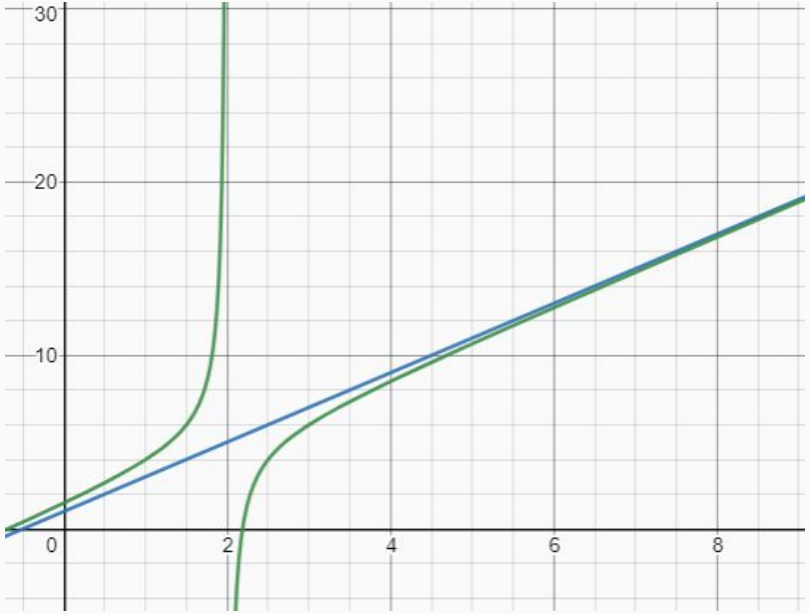
Big-O notation is about ***scalability***, not speed!

Asymptotes

$$f(x) = 2x + 1$$

$$g(x) = \frac{2x^2 - 3x - 3}{x - 2}$$



Big-O notation

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More often you’ll see it (technically, incorrectly) written as:

$$\text{“}f(n) = O(g(n))\text{”}$$

Big-O notation

Definition: $f(\boldsymbol{x}) = O(g(\boldsymbol{x}))$, i.e., "$f(\boldsymbol{x})$ is order *at most* $g(\boldsymbol{x})$", if:

There ***exist*** numbers $\boldsymbol{C}, \boldsymbol{K} > 0$ so that

$$f(\boldsymbol{x}) \leq \boldsymbol{C} \cdot g(\boldsymbol{x}),$$

for every $\boldsymbol{x} \geq \boldsymbol{K}$.

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C: use it to hide the coefficient
K: only care about "big enough" ***x***

Big-O notation

Some examples:

- $3\boldsymbol{x} + 4 = O(\boldsymbol{x})$
- $4\boldsymbol{x}^2 + 10\boldsymbol{x} - 21 = O(\boldsymbol{x}^2)$
- $81\boldsymbol{x} = O(\boldsymbol{x}^3)$
- $\boldsymbol{x}^2 \neq O(\boldsymbol{x})$
- $\log_2 \boldsymbol{x} = O(\ln \boldsymbol{x})$ [base doesn't matter]
- $\boldsymbol{x}^{100} = O(2^{\boldsymbol{x}})$
- $2^{\boldsymbol{n}} = O(\boldsymbol{n}!)$

A linear example

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1. “Guess” a $\boldsymbol{C}$ bigger than 3: try $\boldsymbol{C} = 4$
2. Solve for $\boldsymbol{K}$:

$$3\boldsymbol{n} + 4 \leq 4\boldsymbol{x} \rightarrow \boldsymbol{x} \geq 4, \text{ so } \boldsymbol{K} \text{ can be } 4.$$

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In general, many pairs of *C* and *K* can work.

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1. Guess $\boldsymbol{C} = 5$.
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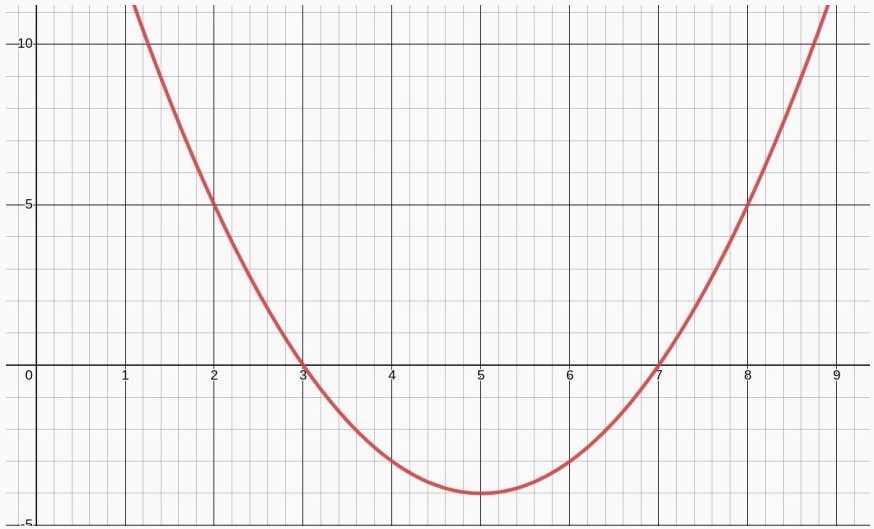
$$4\boldsymbol{x}^2 + 10\boldsymbol{x} - 21 \leq 5\boldsymbol{x}^2 \rightarrow$$

$$0 \leq \boldsymbol{x}^2 - 10\boldsymbol{x} + 21 = (\boldsymbol{x} - 3)(\boldsymbol{x} - 7)$$



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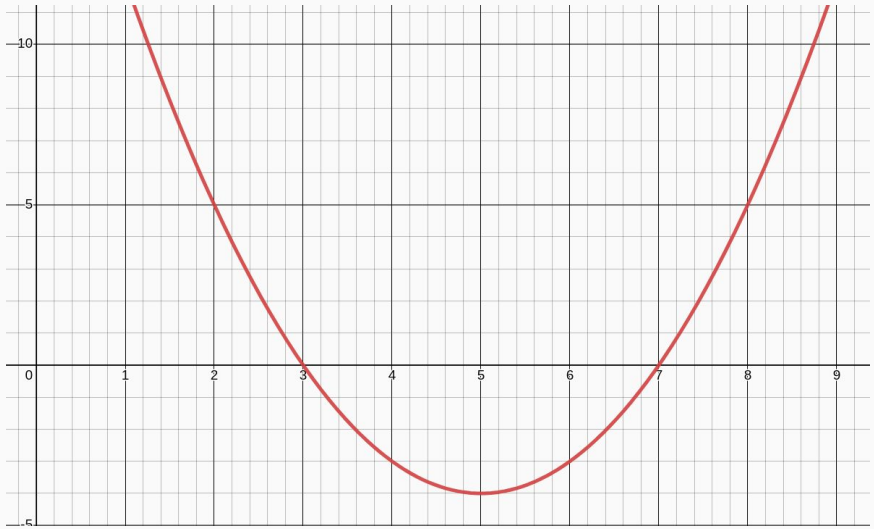
$\boldsymbol{K} = 7$ works, but not $\boldsymbol{K} = 3$ (why?)



Logarithms

Logarithms under different bases are related by the ***change of base*** formula:

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(unless it appears as an exponent)

General themes

For polynomials, look at the highest power, drop the coefficient.

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Simple approach for polynomials

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Simple approach for polynomials

1. Remove negative terms

true for $\boldsymbol{x} \geq 0$

$$3\boldsymbol{x}^5 + 6\boldsymbol{x}^4 - 10\boldsymbol{x}^3 + \boldsymbol{x}^2 - 15\boldsymbol{x} + 11$$

$$\leq \quad 3\boldsymbol{x}^5 + 6\boldsymbol{x}^4 + \boldsymbol{x}^2 + 11$$

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true for $x \geq 0$

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c

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\begin{aligned}
\lim_{x\to\infty} \frac{x^{100}}{e^x} &= \lim_{x\to\infty} \frac{100 \cdot x^{99}}{e^x} \\
&= \lim_{x\to\infty} \frac{100 \cdot x^{99}}{e^x} \\
&= \lim_{x\to\infty} \frac{100 \cdot 99 \cdot x^{98}}{e^x} \\
&\vdots \\
&= \lim_{x\to\infty} \frac{\text{constant}}{e^x} \\
&= 0
\end{aligned}
$$

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For polynomials, look at the highest power, drop the coefficient.

$$203n^5 + 92n^4 - 1423n^3 + n^2 + 86n - 571 = O(n^5)$$

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$$\log_{\boldsymbol{b}} \boldsymbol{n} = O(\boldsymbol{n}^{\boldsymbol{d}}), \text{ for any } \boldsymbol{b} > 1, \boldsymbol{d} > 0$$

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For polynomials, look at the highest power, drop the coefficient.

$$203n^5 + 92n^4 - 1423n^3 + n^2 + 86n - 571 = O(n^5)$$

constant < logarithmic < polynomial < exponential

$\log_b n = O(n^d)$, for any $b > 1$, $d > 0$

$n^d = O(c^n)$ for any $d > 0$, $c > 1$.

Useful properties

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If $f(n) = O(g(n))$, and $h(n) > 0$, then $f(n)h(n) = O(g(n)h(n))$

$n = O(n \log n)$, $n \log n = O(n^2)$, $O(n^2) = O(n^3)$, etc.

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- If $a(\boldsymbol{n}) = O(f(\boldsymbol{n}))$ and $b(\boldsymbol{n}) = O(g(\boldsymbol{n}))$, then $a(\boldsymbol{n}) + b(\boldsymbol{n}) = O(f(\boldsymbol{n}) + g(\boldsymbol{n}))$

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"$O(f(\boldsymbol{n}))O(g(\boldsymbol{n})) = O(f(\boldsymbol{n})g(\boldsymbol{n}))$"

- If $a(\boldsymbol{n}) = O(f(\boldsymbol{n}))$ and $b(\boldsymbol{n}) = O(g(\boldsymbol{n}))$, then $a(\boldsymbol{n})b(\boldsymbol{n}) = O(f(\boldsymbol{n})g(\boldsymbol{n}))$

Useful when considering the complexity of two distinct steps in an algorithm.

Useful properties

- $O(f(\boldsymbol{n})) + O(g(\boldsymbol{n})) = O(f(\boldsymbol{n}) + g(\boldsymbol{n}))$
- $O(f(\boldsymbol{n}))O(g(\boldsymbol{n})) = O(f(\boldsymbol{n})g(\boldsymbol{n}))$
- $O(\boldsymbol{a} \cdot f(\boldsymbol{n})) = O(f(\boldsymbol{n}))$, for constant $\boldsymbol{a} > 0$.
- $f(\boldsymbol{n}) = O(g(\boldsymbol{n}))$, $g(\boldsymbol{n}) = O(h(\boldsymbol{n})) \rightarrow f(\boldsymbol{n}) = O(h(\boldsymbol{n}))$.
- $O(f(\boldsymbol{n})+g(\boldsymbol{n})) = O(\max(f(\boldsymbol{n}), g(\boldsymbol{n})))$

Simplifying expressions

Give a good big-O estimate of f(***n***) using a simple function

$$f(n) = 20\,(n^2 + n^2 \log_4 n)(4n + 3) + (17 \log_3 n + 19)(n^2 + 2)(\ln n - 3)$$

Simplifying expressions

Give a good big-O estimate of f($\boldsymbol{n}$) using a simple function

$$f(\boldsymbol{n}) = 20\,(\boldsymbol{n}^2 + \boldsymbol{n}^2 \log_4 \boldsymbol{n})(4\boldsymbol{n} + 3) + (17 \log_3 \boldsymbol{n} + 19)(\boldsymbol{n}^2 + 2)(\ln \boldsymbol{n} - 3)$$

$$= O(\boldsymbol{n}^2 \log \boldsymbol{n})O(\boldsymbol{n}) + O(\log \boldsymbol{n})O(\boldsymbol{n}^2)O(\log \boldsymbol{n})$$

$$= O(\boldsymbol{n}^3 \log \boldsymbol{n}) + O(\boldsymbol{n}^2(\log \boldsymbol{n})^2)$$

$$= O(\boldsymbol{n}^3 \log \boldsymbol{n})$$

It’s all Greek to me

$f(\boldsymbol{x}) = \Omega(g(\boldsymbol{x}))$

if there are numbers $\boldsymbol{C}, \boldsymbol{K} > 0$ so that

$$f(\boldsymbol{x}) \geq \boldsymbol{C} \cdot g(\boldsymbol{x}), \text{ whenever } \boldsymbol{x} \geq \boldsymbol{K}.$$

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(could also think of as $g = O(f)$)

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$f(\boldsymbol{x}) = \Theta(g(\boldsymbol{x}))$

if there are numbers $\boldsymbol{C}_1, \boldsymbol{C}_2, \boldsymbol{K} > 0$ so that

$$\boldsymbol{C}_1 g(\boldsymbol{x}) \leq f(\boldsymbol{x}) \leq \boldsymbol{C}_2 g(\boldsymbol{x}), \text{ whenever } \boldsymbol{x} \geq \boldsymbol{K}.$$

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(can also think of as $f = O(g)$ and $f = \Omega(g)$)

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Let $f(\boldsymbol{x}) = \boldsymbol{x} \ln \boldsymbol{x}$, $g(\boldsymbol{x}) = \boldsymbol{x}(\boldsymbol{x} + 1)/2$, $\boldsymbol{h}(x) = (\boldsymbol{x} - 1)^3$

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- $f(\boldsymbol{x}) = O(\boldsymbol{x}^2)$, $g(\boldsymbol{x}) = O(\boldsymbol{x}^2)$, $h(\boldsymbol{x}) \neq O(\boldsymbol{x}^2)$
- $f(\boldsymbol{x}) \neq \Omega(\boldsymbol{x}^2)$, $g(\boldsymbol{x}) = \Omega(\boldsymbol{x}^2)$, $h(\boldsymbol{x}) = \Omega(\boldsymbol{x}^2)$

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- $f(\boldsymbol{x}) \neq \Omega(\boldsymbol{x}^2)$, $g(\boldsymbol{x}) = \Omega(\boldsymbol{x}^2)$, $h(\boldsymbol{x}) = \Omega(\boldsymbol{x}^2)$

Putting these together, we get:

- $f(\boldsymbol{x}) \neq \Theta(\boldsymbol{x}^2)$, $g(\boldsymbol{x}) = \Theta(\boldsymbol{x}^2)$, $h(\boldsymbol{x}) \neq \Theta(\boldsymbol{x}^2)$

Only $g(\boldsymbol{x})$ has quadratic growth.

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O: “≤”

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Concluding Remarks

Big-O notation is useful for measuring the **scalability** of **algorithms**, independent of any implementation…

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...but it doesn't always tell the whole story because of the **hidden constant**.

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Not a substitute for benchmarking!

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"Unfortunately, for any instance [graph] that one could fit into the known universe, one would easily prefer $\boldsymbol{n}^{70}$ to even ***constant*** time, if that constant had to be one of Robertson and Seymour's."

-David Johnson, *The NP-Completeness Column: An Ongoing Guide* (1987).